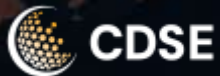


EARTH OBSERVATION SUMMER SCHOOL 2026

- Istanbul Technical University
- 17-21 August 2026

How to become an EO Coder
using AI: Sentinel Hub custom
scripts and CDSE Jupyter Lab

András Zlinszky PhD
Community Evangelist
Copernicus Data Space Ecosystem





Why become an EO coder at all? We have AI, right?

EO Coding with AI is like

Parkour Think of the risks before you jump

Dancing Learn the rules and then you can improvise

Bouldering Hold on fast before your next move

Lego Build from the bricks and ideas will come

Yoga You can only stretch so far. Find your edge

Cycling Gets you faster from Point A to Point B



“Risflecting” –a concept from extreme sports

“Risflecting” = “Risk” + “Reflecting” → thinking about how dangerous your plan is

In AI coding, what is the risk?





Creating a monster

“Risflecting” = “Risk” + “Reflecting” → thinking about how dangerous your plan is

In AI coding, what is the risk?

The risk is creating something you can’t control

Are you creating something that you don’t understand?

1: I sort of understand

2: I understand

3: I could write

4

5: I could



A Calvin and Hobbes Collection by Bill Watterson
Foreword by Pat Oliphant

he basic building blocks

e if I read it

bit of trial and error

myself

same code



You can only stretch yourself so far. Find your edge

The larger the step along the ladder that AI does for you, the more risk you are taking.

So good coders/EO scientists can still write safer code with AI than poor coders/scientists

It is worth becoming a good EO coder

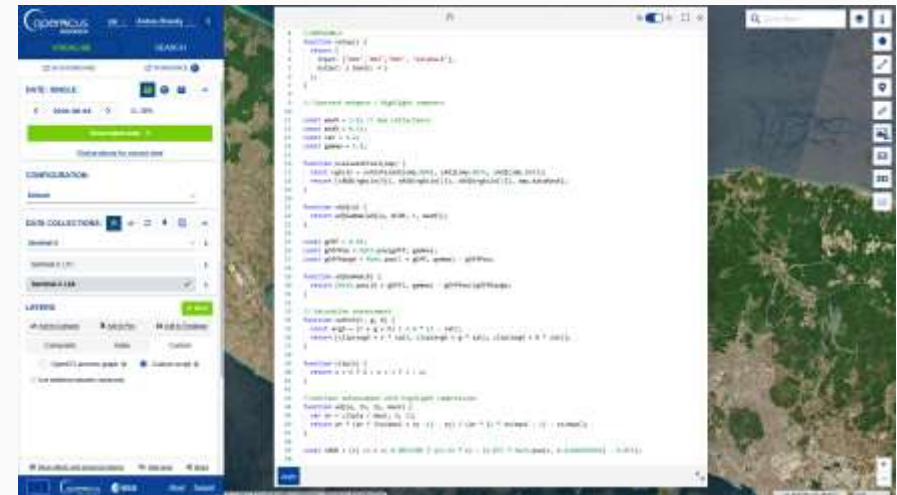
AI can not only write your code, it can also *explain* it



Getting started: working with images in Copernicus Data Space Ecosystem

Copernicus Browser – browser.dataspace.copernicus.eu

The evalscript – the core of analysis and visualization



Key parts of the evalscript

Setup

Defining the input
and output
parameters

- Default [RGB]
- Index [Float32]
- Eobrowserstats [Float32]
- Datamask [Boolean]

```
1 //VERSION=3
2 let minVal = 0.0;
3 let maxVal = 0.4;
4
5 let viz = new HighlightCompressVisualizer(minVal, maxVal);
6
7 function setup(ds) {
8   return {
9     input: ["B03", "B04", "B08", "dataMask"],
10    output: { bands: 4 }
11  };
12 }
13
14 function evaluatePixel(samples) {
15   let val = [samples.B08, samples.B04, samples.B03, samples.
dataMask];
```

Evaluatepixel

The actual
mathematical
operation

- Composite
- Spectral index
- Regression
- Classification

Optional parts of the evalscript

User variables

Defining the
settings for
visualization (or
calculation)

Typically done in
the header if you
want a clean job

```
1 //VERSION=3
2 let minVal = 0.0;
3 let maxVal = 0.4;
4
5 let viz = new HighlightCompressVisualizer(minVal, maxVal);
6
7 function setup(ds) {
8   return {
9     input: ["B03", "B04", "B08", "dataMask"],
10    output: { bands: 4 }
11  };
12 }
13
14 function evaluatePixel(samples) {
15   let val = [samples.B08, samples.B04, samples.B03, samples.
dataMask];
```

Utility function

Commands for
regularly used
visualization tools

ColorRampVisualizer
CreateRedTemperature
etc.

[https://documentation.d
ataspace.copernicus.e
u/APIs/SentinelHub/Ev
alscript/Utilities.html](https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Evalscript/Utilities.html)

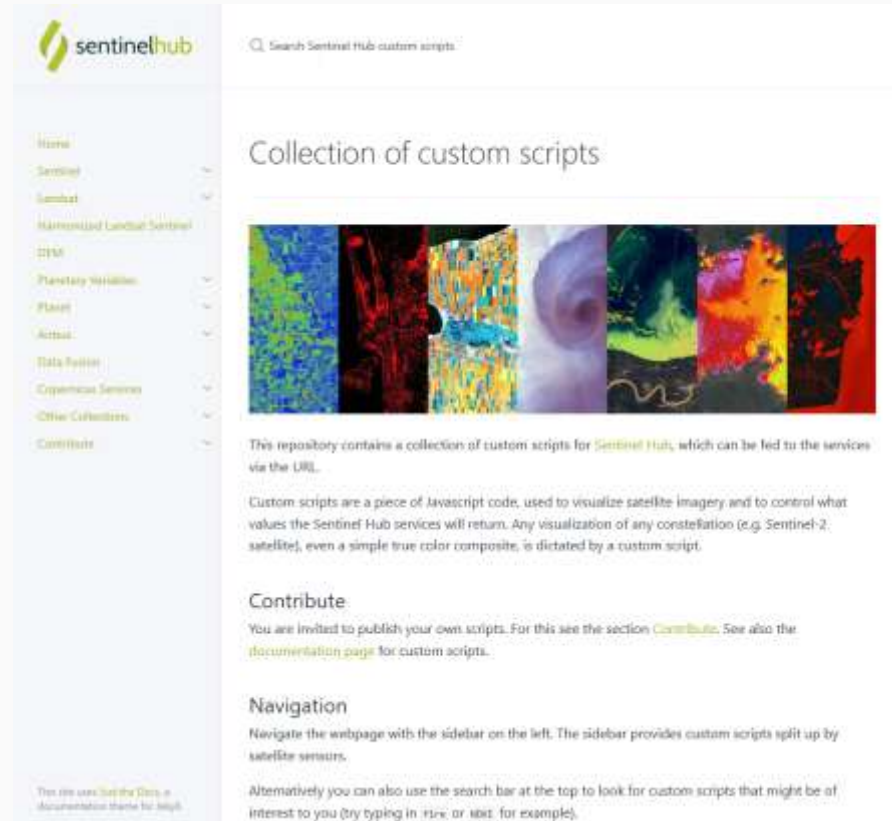
The custom script repository: custom-scripts.sentinel-hub.com

Collection of evalscripts for various satellites and use cases

Organized by satellite and application fields

Most of them user contributed → so quality is not consistent...

Alternative (for CLMS, Landsat etc <https://github.com/eu-cdse/sentinel-hub-custom-scripts>)



Collection of building blocks for new custom scripts

+ agents.md
a document that helps AI tools write correct contributions

https://github.com/sentinel-hub/custom-scripts/blob/main/AGENT_S.md

Copy this to your working directory if you are coding locally

Hands-on block 1: Exploring Copernicus Browser and evalscripts

1. [Browser.dataspace.copernicus.eu](https://browser.dataspace.copernicus.eu)
2. Zoom/pan to a location you know well
3. Load satellite images and/or CLMS data
4. Explore the available data layers
5. Click “Custom” and explore the drag-and drop custom scripting interface
6. Click `</>` to open the code window and select “Custom Script”
7. Try editing code manually, change settings of scripts and see what happens
8. Select a custom script from the repo, copy and paste into the code window, apply



Hands-on block 1: Starting to work with AI – free tools

Quiz: which custom geospatial is this?

Ai Assistant	Registration required	Free tier prompt limits and quotas
Gemini	Yes (Google Account)	50-100 daily credits
ChatGPT	No (Guest mode available)	Dynamic rolling limits capped based on current server demand
Claude Code	Yes (Email/Google/Apple ID)	15-40 messages every 5 hours
Microsoft Copilot	No (Guest mode available)	Dynamic limits, better if you sign in with free MS account
GitHub Copilot	Yes (GitHub account)	50 chat messages, 2000 code completions monthly

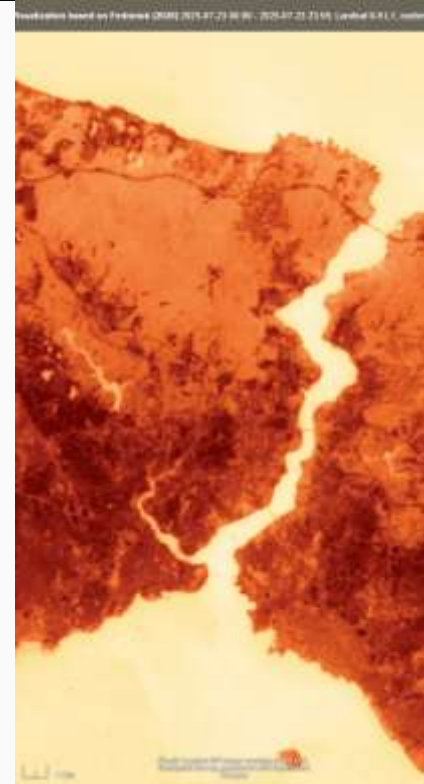
Hands-on block 1: Our first AI prompts

1. Combine custom scripts. Take a spectral index and prompt for a different visualization palette. Eg: *“can you please take the sentinel hub Sentinel 2 SAVI index and apply a different palette to it? It should have user-tuneable min and max values, and a wider range of colours, with the densest vegetation blending towards purple, healthy vegetation in green, unhealthy vegetation in white and very low values in red.”*
2. Read the code **before** you paste it into the code window and think. Will it work?
3. If you encounter a bug, **read the error message**, then paste it to your AI
4. If you have a working script, place it in an online repo. It will be useful later.
5. Bonus task: take the colour requirements laid out here <https://earsc.org/2024/10/23/expandeo-2026-image-contest/> and make an evalscripts that uses them!



Hands-on block 1: Making scripts from papers

1. Take a scientific paper and prompt AI to recreate the analysis in a custom script:
Eg *“can you implement the visualization algorithms proposed in this paper in the form of new Sentinel Hub custom scripts for landsat thermal?”*
<https://www.earthdoc.org/content/papers/10.3997/2214-4609.2024510007?crawler=true>
2. Check the paper first – what do you think, will it work?
3. Find the exact mathematical operation, describe it in an equation or a few sentences for yourself – check against the finished script
4. Bonus task: can you create a script from this paper? Jiang 2020 ISPRS: a new spectral index for identifying water <https://doi.org/10.5194/isprs-annals-v-3-2020-33-2020>



Hands-on block 1: Shortening, adapting to other datasets

1. Only rather short scripts can be used for generating sharing links/url-s
2. Take a really complex script and ask AI to shorten/minify, Eg *“can you create a minified version of the Pierre Markuse Sentinel-2 wildfire custom script? It should run in Copernicus Browser and have exactly the same functionality (including tuning options), but the code should be as short as possible. ”*
3. Read the code – if you don’t understand, ask AI to explain
4. Take an evalscripts and ask AI to adapt to a different dataset: Eg. *“can you make me a script for Sentinel-2 quarterly mosaics based on the Harel Dan agri growth stage script here https://custom-scripts.sentinel-hub.com/custom-scripts/sentinel-2/agriculture_growth_stage/?”*
5. Bonus task: try both – a different dataset, then shortening!

Cuando river, Zambia-
Angola-Namibia border,
Agri Growth Stage on Sen-
2 quarterly mosaics



CDSE

EARTH OBSERVATION SUMMER SCHOOL 2026

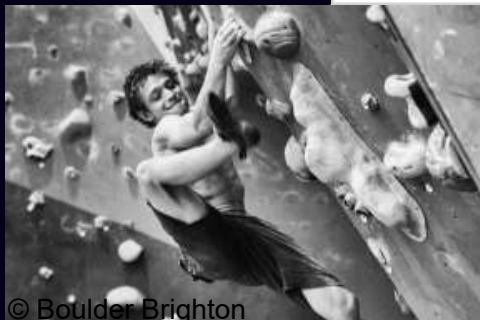
● Istanbul Technical University
● 17-21 August 2026

Hands-on k

1. Build a s
2. Take a s
[Kernel M](#)
3. Combine



Hold on fast before your next move



© Boulder Brighton

What have we learned so far? How can **you** help AI?

- Think before you prompt. Make sure you know what you want (but AI can help planning! "*Is this spectral index relevant for that application? Find me literature!*")
- Specify platform, input data, output data and operation (regression, composite, classification, spectral index) **clearly** in the prompt



Hold on fast before your next move

What have we learned so far? How can **you** help AI?

- Always read the code AI creates for you. If you don't understand part of it, ask AI to explain. Don't be shy!



• This is the essential difference
between AI making you a better coder
or a worse coder

- Always read error messages. Don't just copy! If you have an idea what the problem could be, let AI know.



Hold on fast before your next move

What have we learned so far? How can **you** help AI?

can you create me a Sentinel Hub custom script that produces (what looks like) variable rate fertilization zones? It should be based on the mean kNDVI from the April and July mosaic, categorize NDVI into five bins, and produce a raster map colored according to the amount of applied fertilizer (with realistic values).



- Verify with real-world cases. Use 3-4 independent locations, including one you know well. Compare with other sources of information. AI can help find specific events or conditions (e.g. *“when was there a heatwave in Istanbul?”*)
- Check numeric outputs using histograms, timelapses, time series. Do they look right? How do they compare to other evidence?
- Think about edge cases and negative examples. What conditions would break your script? Where would it give false positives? What is the easiest mistake to make?

Moving beyond Copernicus Browser – the API request and Jupyter

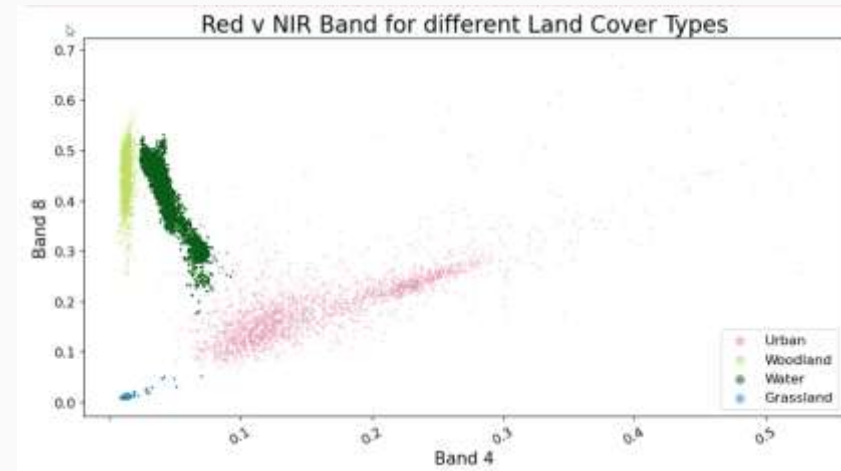
Quiz: which custom script is this?

- Every Sentinel Hub command is an API request – your machine communicates with the server and the data is prepared for you
- This is what a typical Sentinel Hub API request includes
 - Input data [which data collection, which service url]
 - Area of interest, time of interest
 - Data processing/analysis [evalscript]
 - Authentication [configuration]
 - Response type [data format]
- <https://shapps.dataspace.copernicus.eu/requests-builder/>

Moving beyond Copernicus Browser – the API request and Jupyter

Quiz: Which custom script is this?

- Sentinel Hub API requests enable you to embed data analysis functionality in your code, without downloading data
- You can
- Extract coordinates of an object on an image
- Look at a series of images and create event alerts
- Build new quantitative products
- Create interactive visualizations



Coding with AI – in a web browser vs your local coding environment

Quiz: which custom script is this?

	Desktop coding environment	CDSE Jupyter Lab
Performance	Depends on your machine	Depends on server flavor
Data access	API-based but remote	Very fast (runs on the same physical infrastructure the data is stored on)
Maintenance, dependencies	You maintain it	Pre-installed and updated
AI assistant	Can be integrated to your environment	External (browser or desktop)



Learn the rules and then you can improvise

CDSE requests builder and the structure of an API request



CDSE requests builder

Log in Requests Send

Select API

Process Catalog Statistical OGC Services Search Statistics Search Add Query Data Import

Data Collection

Collection 1 From 2009-07-13 To 2009-08-10 Identifier 07000000 Options Data Fusion

Output

Height / Width Resolution Width 512 Height 343.667 Aspect ratio Image Format JPEG Identifier default Add

Area of interest

EPSG:4326 Area 88.81 km² MultiPolygon

12.66693, 41.870072, 12.541001, 41.017096

Request Preview



Learn the rules and then you can improvise

The same in CDSE Jupyter Hub <https://jupyterhub.dataspace.copernicus.eu>



```
File Edit View Run Kernel Help Settings Help

/ ... / sentinelhub / getting_started /
Name Modified
customs_scripts 11 hours ago
statistical_api 4 hours ago
cloudless_processing_scripts 11 hours ago
data_download_processing_request_api.py 2 seconds ago
from_sentinel2_to_jupyter.py 11 hours ago
sentinelhub_to_jupyter.py 11 hours ago
imagery_from_sentinelhub_to_jupyter.py 11 hours ago
scale_to_CDSE.py 11 hours ago

In [ ]:
# Sentinel-2 data download and processing

# The notebook in the example is used to select the appropriate bands. We return the RGB (B04, B03, B02) Sentinel-2
# L2A bands.

# The least cloudy image from the time period is downloaded. Without any additional parameters in the notebook, the
# downloaded data will correspond to reflectance values in [0,100]. Normal values in 0-255 range.

%%writefile true_color.py

def sample():
    return {
        'input': [
            'B04', 'B03', 'B02'
        ],
        'output': [
            'B04', 'B03', 'B02'
        ]
    }

def evaluatePixel(sample):
    return [sample.B04, sample.B03, sample.B02]

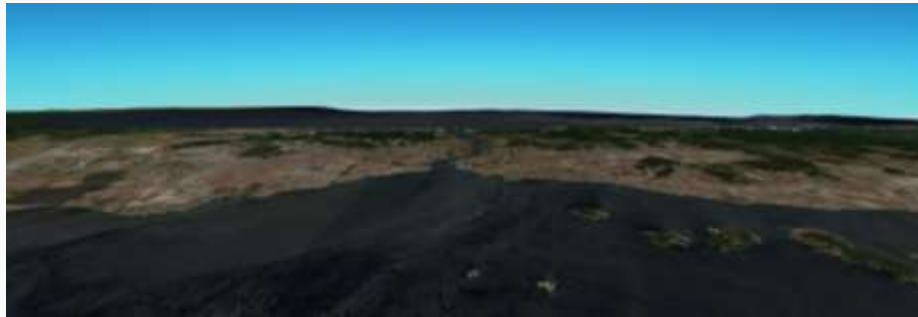
request_true_color = SentinelhubRequest(
    notebook='sentinelhub_to_jupyter.py',
    input_data={
        'Sentinel2L2ARequestInput': {
            'data_collection': 'Sentinel2_L2A',
            'name': 'j11s',
            'service_url': 'https://ds.dataspace.copernicus.eu'
        },
        'time_interval': ('2022-01-01', '2022-01-01'),
        'other_args': {'datafilter': 'maxcloudscore'}
    },
    responses=[SentinelhubRequestOutput, response={'default', 'timeType', 'B04'},
    time_interval,
    time_interval,
    config=config,
    )

The returned 'get_data()' will always return a list of length 3 with the available image from the requested time
```

Hands on block 2: Coding Jupyter notebooks with AI

Quiz: which custom script is this?

- Launch CDSE Jupyter Hub <https://jupyterhub.dataspace.copernicus.eu>
- Create Oauth Client credentials in Sentinel Hub dashboard
<https://shapps.dataspace.copernicus.eu/dashboard/#/account/settings>
- Explore tutorial notebooks – beginners: https://github.com/eu-cdse/notebook-samples/blob/main/sentinelhub/getting_started/from_browser_to_jupyter.ipynb
- Explore tutorial notebooks – if you have coding experience: https://github.com/eu-cdse/notebook-samples/blob/main/sentinelhub/use_cases/air_pollution_statistics.ipynb



Hands on block 2: Coding Jupyter notebooks with AI

Quiz: Which custom skill is this?

- Creating a notebook with AI starts with planning. What dataset will you use? What is the desired outcome? Who is the user group? What algorithms and steps do you plan? What choices should the user make? Prompt AI to create a plan for a notebook! Eg: *“I want to create a Jupyter notebook that classifies Sentinel-2 time series pixels to identify maize crops. It should use a classifier trained on CLMS crop type pixels. Then, it should identify parcels harvested unusually early due to drought.”*
- A good AI assistant will ask you questions to refine the request
- Once you have answered all questions and you are satisfied with the plan, you can prompt AI to build
- Ask AI to create a skill file on the way – **read the skill file** and correct it, point to documentation and examples

Hands on block 2: Setting up CDSE Jupyter Lab as a backend in your local IDE

- Start by installing Jupyter/Jupyter Hub extension (e.g. for VSCode)
- Find command for specifying Jupyter Server URL and choose “Connect to a JupyterHub (not “Connect to a Jupyter Server”)
- When prompted for the URL, enter <https://jupyterhub.dataspace.copernicus.eu/> . Add your CDSE user name
- Log in to JupyterHub, then go to:
<https://jupyterhub.dataspace.copernicus.eu/hub/token> Click "Request new API token" and copy the generated token
- When asked for password/token, paste this token
- Select kernel (Sentinel Hub in this case)

A real world case of an AI-generated notebook – monitoring fishing activity with Sentinel-2 and Global Fishing Watch

- Open https://github.com/eu-cdse/notebook-samples/blob/main/sentinelhub/data_integration/ship_detection_and_ais_identification.ipynb
- Notebook started as a feature for the CDSE gallery, but I was curious to see whether the boats I find are truly fishing boats
- I found out Global Fishing Watch serves ship transponder data through an API and this was compatible with CDSE Jupyter Lab





A re
cont

• |

"I have
attended
a major
color
doma
CDS

Event Categories

- Agriculture (5)
- Defence and Intelligence (1)
- Earth Observation (9)
- Governance and policy (11)
- Hackathon (3)
- Industry (8)
- Natural Sciences (2)
- Software development (4)
- Training Courses (4)

Event Sizes

- 100 attendees
- 1,000 attendees
- 10,000 attendees

Online Events



"I can
Geoc
We c
repre
notel





Build from the bricks and ideas will come

AI and Jupyter can leverage many different data sources, datasets and code tools – creating functioning links between them



- Custom scripts repo
 - Sentinel Hub Statistical API
 - Cartography and visualization libraries (Folium, Matplotlib)
 - Advanced machine learning and computer vision (OpenCV, Scikit-learn)
 - Other data source API-s
-
- But it makes a big difference if you know them too, and make suggestions



Gets you faster from point A to point B

What have we learned?



©Instagram.com/rollingspoke

AI can make users smarter, or it can ruin their skills. It is up to the user.

You don't want to build something you don't understand – and you have to know the risks

CDSE makes data analysis code simple enough that AI can produce good scripts – but verification is always necessary

CDSE Jupyter Lab enables the user to combine a multitude of datasets, tools and analysis methods in a reasonable amount of time

CDSE is ready for the open source sharing ecosystem

Open your head – but not so much your brain
falls out!

Earth Observation
Summer School 2026

● Istanbul Technical University
● 17-21 August 2026

Useful resources

- [Documentation.dataspace.copernicus.eu](https://documentation.dataspace.copernicus.eu)
- <https://www.youtube.com/@copernicusdataspaceecosystem>
- <https://github.com/eu-cdse/notebook-samples/tree/main/sentinelhub>
- CDSE email newsletter:
<https://dataspace.copernicus.eu/webform/newsletter>